# Prodigy Anesthesia Quick Review

## Spinal Cord Injury

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| What is the most effective device for minimizing cervical motion? | Halo-thoracic immobilization devices.<br><br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 256.* |
| What is the median age of the patient who suffers a spinal cord injury? | 25<br><br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 924.* |
| What is the leading cause of death in patients with spinal cord injury at the scene of the injury? | respiratory failure from muscle weakness and aspiration<br><br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 924.* |
| Fibers from what nerve roots innervate the diaphragm? | C3, C4, C5<br><br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 258.* |
| What is the incidence of cervical spine injury in all major trauma victims? | Between 1.5% and 3% of all major trauma victims have a cervical spine injury<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 255.* |
| The likelihood of a cervical spine injury is considered minimal if what five criteria are met? | 1) No tenderness over the midline cervical spine area, 2) no focal neurologic deficits, 3) normal sensorium, 4) no intoxication with drugs or alcohol, and 5) no other painful injury that could distract from the patient's evaluation of symptoms.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 255-256.* |
| What are the symptoms of acute spinal cord injury? | Below the level of injury, the patient will exhibit flaccid paralysis, total absence of sensation, and loss of temperature regulation and spinal cord reflexes.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 255.* |
| What are the symptoms of spinal shock? | Decreased blood pressure due to loss of preload (there is significant dilation of the capacitance vessels). If the injury is at or above T1-T4, the loss of sympathetic innervation to the heart can result in bradycardia.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 257-258.* |

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| How does the skin of patients with spinal shock compare to that of patients with hemorrhagic shock? | Because spinal shock disrupts the compensatory vasoconstrictive response, patients with spinal shock will exhibit warm, pink extremities whereas patients with hemorrhagic shock will exhibit cool, clammy extremities.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 926.* |
| What monitoring techniques should be instituted for patients with spinal shock? | In addition to standard monitors, an arterial line should be placed to allow for moment-to-moment observation of blood pressure as well as ease-of-access for arterial blood gas analysis. A pulmonary artery catheter may also be used to facilitate management of fluid and drug therapies.<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1507.* |
| What is autonomic hyperreflexia? | Autonomic hyperreflexia may appear after spinal shock has resolved and represents the return of spinal cord reflexes. It is a massive sympathetic discharge that occurs in response to a cutaneous or visceral stimulation below the level of transection. In normal patients, the sympathetic activity that results from this stimulation is overridden by inhibitory impulses from higher central nervous system centers. In the spinal cord injured patient, the sympathetic outflow is isolated from the inhibitory feedback loop and the sympathetic discharge is unopposed.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259.* |
| With what spinal cord injury levels is autonomic hyperreflexia most common? | About 85% of patients with a spinal cord injury above T6 will exhibit autonomic hyperreflexia, but it is uncommon in patients with lesions below T10.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259.* |
| What are the signs of autonomic hyperreflexia? | Hypertension, bradycardia, and dysrhythmias.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 927.* |
| What types of stimuli may precipitate autonomic hyperreflexia? | Bladder catheterization, defecation, childbirth, distention of the bladder or rectum, and cutaneous stimulation.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 927.* |
| What precautions can you take to prevent or treat autonomic hyperreflexia during the perioperative period? | Don't allow stimulation from surgical incision or other manipulation to occur before adequate general or regional anesthesia has been established. Have immediate access to short-acting vasodilators such as nitroprusside to treat sudden, severe hypertension. Monitor the patient closely in the recovery area as autonomic hyperreflexia may not occur until the effects of anesthetic drugs wear off.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259-260.* |

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| How can autonomic hyperreflexia be prevented in a parturient with a chronic spinal cord injury undergoing childbirth? | The use of neuraxial anesthesia has been reported to prevent autonomic hyperreflexia in response to labor contractions. Epidural anesthesia is less effective than spinal anesthesia in this scenario as it can spare sacral segments that can be involved in autonomic hyperreflexia.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 260.* |
| Will topical anesthesia prevent autonomic hyperreflexia from occurring in patients undergoing bladder procedures such as cystoscopy? Why or why not? | No. Topical anesthesia does not affect the nerves that detect bladder distention which can precipitate autonomic hyperreflexia.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 260.* |
| How does the level of the spinal cord injury affect the patient's pulmonary status? | If the level of injury is above C5 then the patient will likely be apneic without mechanical ventilation. Lower cervical and thoracic transections can leave the accessory muscles of ventilation paralyzed resulting in a decreased ability to ventilate properly or clear secretions.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 258.* |
| What are the gastrointestinal complications associated with high spinal cord injury? | The loss of sympathetic tone can predispose patients to peptic ulceration, especially when combined with the administration of corticosteroids. Orogastric or nasogastric tubes are often employed to decompress the stomach which has the combined effect of easing restrictions on diaphragmatic movement caused by gastric distention and reducing the risk for aspiration of gastric contents.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 926.* |
| What are the urinary complications that may arise from chronic spinal cord injury? | Bladder infections become common as the patient is unable to empty the bladder completely. This may also predispose the patient to renal stone formation and even progress to renal failure, a common cause of death in patients with chronic spinal cord injury.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 258.* |
| How do patients with acute and chronic spinal cord injury respond to succinylcholine? | Because of the risk for severe hyperkalemia following the administration of succinylcholine, especially during the first six months after the injury, it should be avoided in all patients with spinal cord injury for the rest of their life. Some practitioners will utilize succinylcholine during the first 24 hours as upregulation of extrajunctional receptors probably has not had time to occur yet.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259.* |

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| What are the complications that may arise from prolonged immobility in a patient with chronic spinal cord injury? | The chronic immobility predisposes the patient to osteoporosis, atrophy of skeletal muscles, and an increased risk for venous thrombosis.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 258.* |
| Why is it important to closely manage hydration status in a patient with spinal cord injury? | Although they may have difficulty maintaining ventricular filling pressures due to the loss of venous return from the cord transection, they are prone to pulmonary edema if overhydration occurs.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 257.* |
| How does chronic pain arise from chronic spinal cord injury? | Nerve root pain often presents at or near the site of the original injury. Phantom pain may occur in areas of complete denervation.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 258.* |
| How long does spinal shock last? | In patients who survive, spinal shock can last 1-3 weeks.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 255.* |
| How effective are hard cervical immobilization collars at restricting movement of the neck? | Hard cervical collars only reduce neck motion by about 25%.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 256.* |
| What is the recommended technique for limiting movement of the neck during laryngoscopy when a cervical injury is suspected? | Laryngoscopy should be performed with an assistant who provides manual in-line stabilization.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 256-257.* |
| What is the annual incidence of spinal cord injury? | Approximately 10,000 spinal cord injuries occur each year.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 924.* |
| How is manual in-line stabilization performed? | The person performing manual in-line stabilization rests both hands on each side of the patient's face with the tips of the fingers resting on the mastoid processes. A downward pressure is applied to keep the head still against the table and prevent flexion of extension of the neck during laryngoscopy.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 256-257.* |
| What vertebral body is most commonly injured in cervical spine fractures? | C7<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 925.* |

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| Is an LMA suitable for the management of a patient with a suspected cervical spine injury who is apneic or unconscious and hypoventilating? | An LMA or Combitube is suitable for placement by paramedics for transport, but should be replaced with a definitive endotracheal airway as soon as possible.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 925.* |
| Would the use of a McCoy laryngoscope be appropriate for securing the airway in a patient undergoing evaluation for cervical spine injury? What about a flexible fiberoptic intubation? Translaryngeal guided intubation? | While direct laryngoscopy with a McCoy laryngoscope enables intubation with minimal neck movement, flexible fiberoptic intubation and translaryngeal guided (retrograde) intubation enable the anesthetist to secure the airway with no movement of the neck which is crucial in patients with cervical spine injury.<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1494.* |

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